

PLATTSMOUTH, NE, USA

WMO: 722291

Lat: 40.948N

Lon: 95.917W

Elev: 1205

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 00470

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	3.1	-9.5	3.5	2.0	-6.1	4.1	5.3	30.9	26.8	27.8	25.2	8.7	320	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	93.3	75.2	90.6	75.0	88.2	74.0	79.6	88.6	77.6	86.4	75.9	84.9	10.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	147.2	86.1	74.9	136.8	83.6	73.0	128.2	81.5	44.1	88.1	42.0	86.9	40.2	85.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	22.5	19.4	-9.1	97.7	4.4	2.3	-12.3	99.4	-14.9	100.7	-17.3	102.0	-20.5	103.6		
			-9.7	82.2	4.3	1.5	-12.8	83.3	-15.4	84.2	-17.8	85.1	-20.9	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	24.8	27.3	41.9	52.3	64.0	73.7	76.6	74.2	67.6	54.4	41.3	29.1
	DBStd	20.67	11.80	12.31	12.26	10.33	8.85	5.94	5.56	5.35	7.80	9.11	10.57	10.71
	HDD50	2834	780	641	311	96	7	0	0	0	1	58	292	648
	HDD65	5784	1245	1057	721	395	129	5	1	3	62	343	710	1113
	CDD50	3712	0	4	59	165	441	710	825	750	529	196	32	1
	CDD65	1186	0	0	4	13	98	266	361	288	140	16	0	0
	CDH74	10902	0	0	52	174	896	2494	3540	2426	1164	151	5	0
	CDH80	4003	0	0	9	47	297	932	1429	879	379	31	0	0
Wind	WSAvg	8.9	9.9	9.8	10.6	11.2	9.5	8.2	6.3	6.2	7.4	8.8	9.9	9.4
Precipitation	PrecAvg	31.9	0.8	1.1	1.9	3.3	5.1	4.6	4.5	3.2	2.8	2.3	1.6	0.9
	PrecMax	51.7	1.6	3.3	4.6	7.3	9.6	8.3	16.1	8.3	6.3	7.4	3.3	2.4
	PrecMin	23.3	0.1	0.2	0.0	0.6	1.4	1.3	1.0	0.6	0.6	0.0	0.0	0.0
	PrecStd	6.7	0.4	0.8	1.3	1.5	2.6	2.0	3.3	2.2	1.6	1.7	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.2	66.7	80.8	86.0	93.3	95.1	98.6	95.5	92.5	83.7	73.2	58.8
		MCWB	43.7	50.9	61.1	63.9	68.8	74.8	74.2	75.8	73.0	67.9	55.9	48.2
	2%	DB	50.0	56.6	73.0	79.2	87.0	91.3	93.3	91.1	88.4	78.5	67.6	53.0
		MCWB	42.2	47.3	55.2	59.8	66.8	75.0	76.8	75.6	72.8	62.0	55.2	45.6
	5%	DB	44.9	50.1	67.0	73.3	82.5	88.5	90.6	88.3	84.5	73.4	61.9	48.2
		MCWB	38.6	42.2	53.4	57.0	64.9	73.8	76.8	75.3	71.0	59.7	51.3	42.2
	10%	DB	40.5	43.9	61.1	69.4	79.1	85.8	87.8	85.0	80.9	69.9	56.8	44.2
		MCWB	36.5	38.4	50.8	55.3	63.8	72.0	75.3	73.4	68.5	57.6	47.8	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	53.8	63.7	65.5	73.4	80.9	82.3	81.3	77.1	71.3	60.3	53.0
		MCDB	53.3	59.6	72.1	80.1	83.0	90.3	91.2	90.6	87.1	79.6	67.8	54.9
	2%	WB	42.2	46.5	59.5	62.5	70.6	78.0	80.0	78.7	75.0	66.5	56.0	46.5
		MCDB	48.7	56.7	68.2	74.2	81.5	87.4	89.3	86.8	84.4	73.4	64.3	50.4
	5%	WB	39.3	42.3	55.1	60.1	68.3	75.8	78.0	76.8	73.3	62.8	52.2	42.6
		MCDB	44.2	49.4	64.0	70.2	79.0	85.8	86.8	85.0	80.7	69.9	60.9	48.0
	10%	WB	36.4	38.7	50.5	57.5	66.3	74.0	76.1	74.9	71.3	59.7	48.5	39.2
		MCDB	40.7	43.9	61.7	66.5	75.9	83.2	85.0	82.9	78.5	67.3	56.7	43.7
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.0	20.1	20.7	19.4	18.5	18.1	18.4	20.4	20.1	18.8	15.6
		MCDBR	22.8	25.1	29.7	27.8	24.2	21.1	20.4	21.6	23.1	25.9	25.9	22.6
		MCWBR	15.8	16.0	15.4	13.8	9.9	9.8	9.4	10.0	10.2	13.0	14.2	16.1
	5% WB	MCDBR	20.8	23.2	25.0	23.9	20.6	19.3	18.1	18.7	19.5	20.8	23.8	20.9
MCWBR		14.8	15.8	14.4	13.7	9.8	10.7	10.1	10.3	9.9	13.2	15.1	15.8	
Clear-Sky Solar Irradiance	taub	0.278	0.314	0.325	0.380	0.407	0.412	0.422	0.414	0.373	0.331	0.297	0.282	
	taud	2.471	2.383	2.413	2.277	2.270	2.308	2.276	2.308	2.378	2.485	2.506	2.467	
	Ebn at Noon	283	285	294	283	276	274	270	269	274	276	272	271	
	Edh at Noon	24	30	33	41	42	41	42	39	34	27	23	22	
All-Sky Solar Radiation	RadAvg	652	943	1225	1557	1845	2053	2061	1791	1479	1039	726	549	
	RadStd	54	58	110	104	145	156	137	105	97	119	53	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page